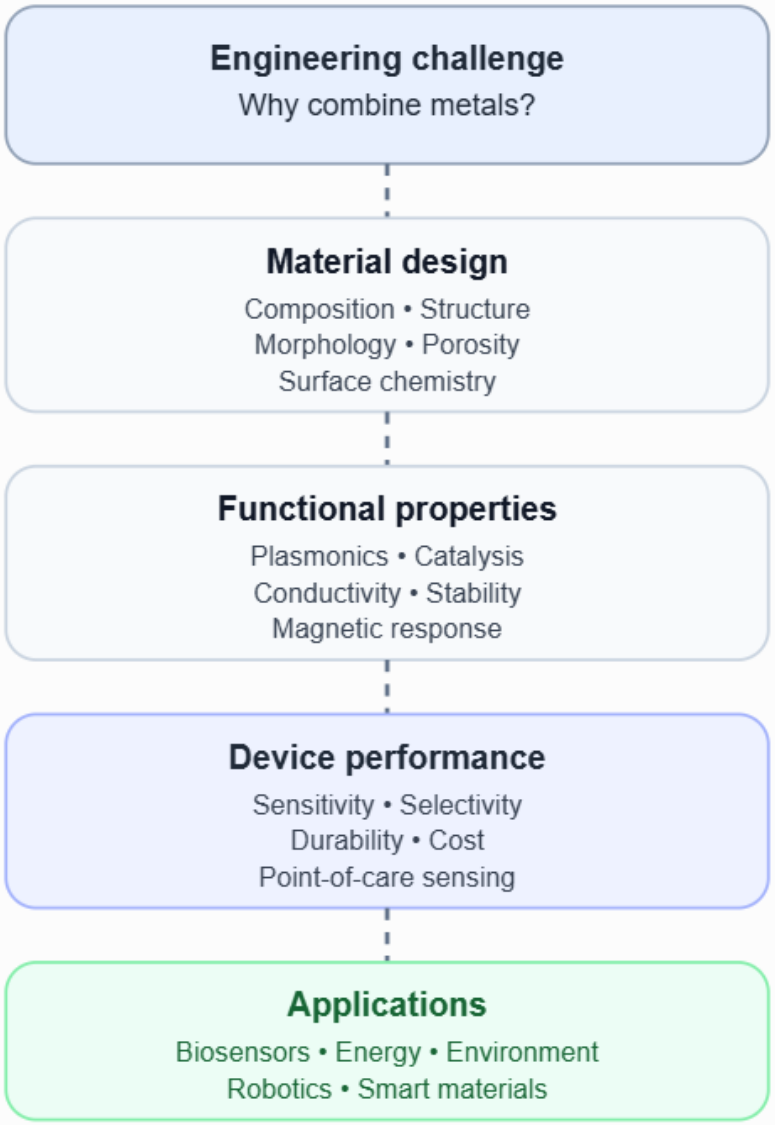
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AUSTRALIA
CREATE CHANGE

Bi- and Trimetallic Nanocomposite for Biosensing

Mostafa Kamal Masud
Next Generation Cancer Research Fellow II ARC DECRA Recipient (2024)
Australian Institute for Bioengineering and Nanotechnology (AIBN), The University of Queensland, Brisbane, QLD 4072, Australia.
Email: m.masud@uq.edu.au

[MATE6301] Nanomaterials
School of Mechanical and Mining Engineering
The University of Queensland

Why do engineers design multimetallic nanomaterials?



What will we Learn?

- ❑ Why bimetallic and trimetallic nanomaterials exhibit synergistic properties.
- ❑ Compare alloy, core–shell and porous multimetallic structures.
- ❑ Describe how composition and morphology influence optical sensing performance.
- ❑ Evaluate fabrication strategies for mesoporous multimetallic materials.
- ❑ Propose a nanocomposite design for an engineering sensing application.

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Why they are important?

Synergistic Effects: Alloying metals produces new electronic, catalytic, and mechanical properties that outperform single-metal systems.

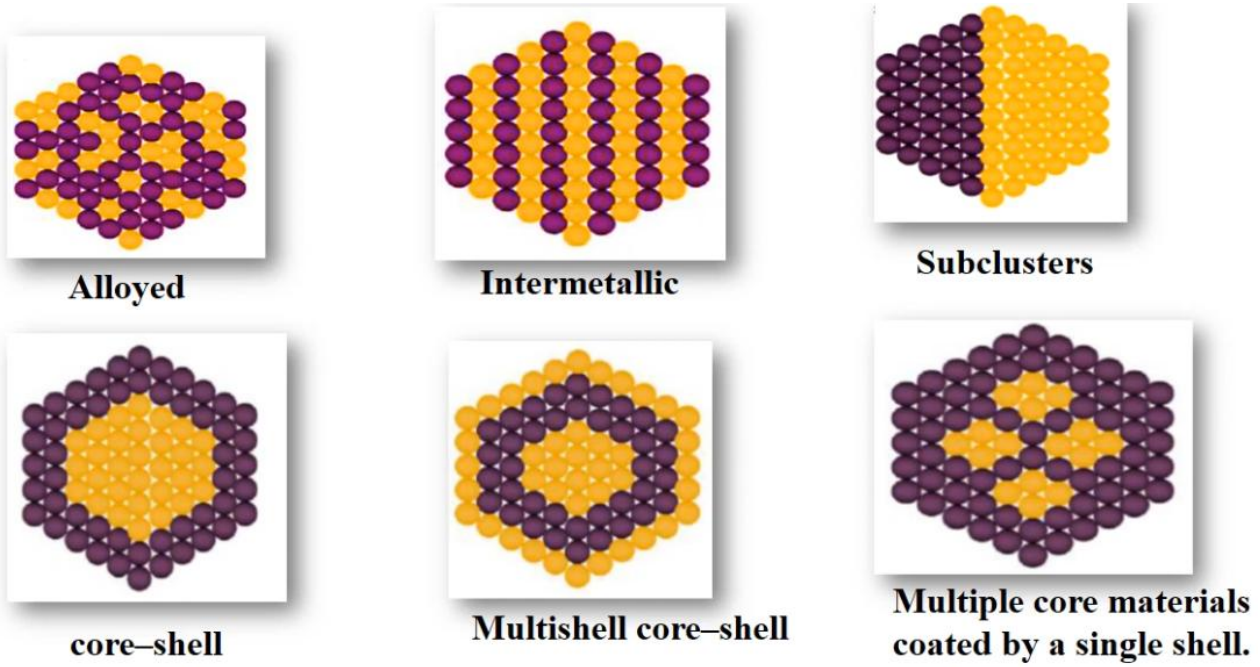
Tunable Properties: Engineers can adjust metal type, ratio, and structure to fine-tune performance for targeted applications (e.g., sensors, catalysts, coatings).

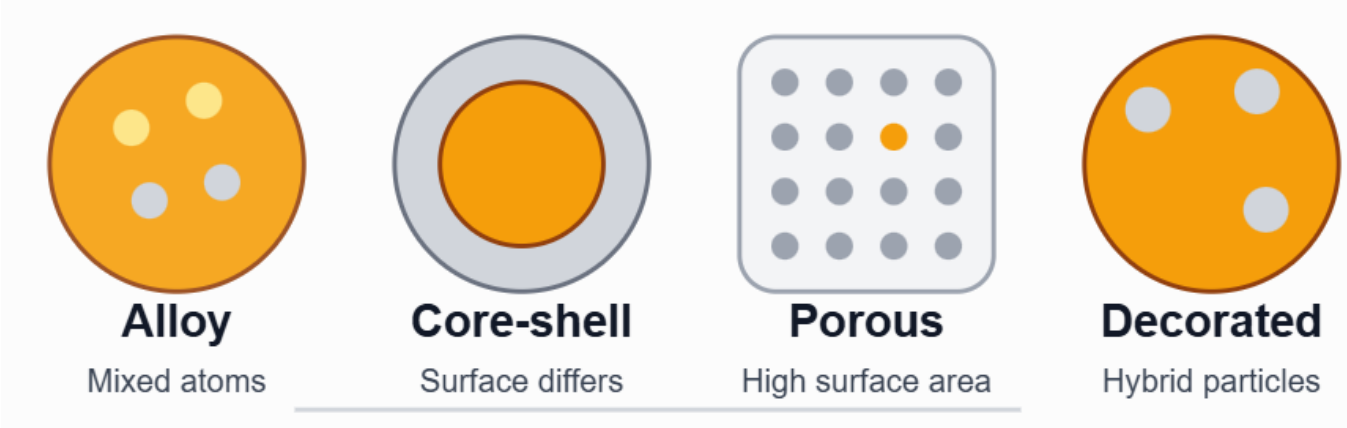
Enhanced Performance: Trimetallic systems can improve catalytic activity, stability, and selectivity compared to bi-metallic and monometallic nanoparticles, enabling more efficient and durable designs.

Design parameter	Engineering consequence
Metal composition	Electronic band structure changes
Atomic arrangement	Catalytic activity changes
Nanostructure geometry	Optical field enhancement
Porosity	Surface area and mass transport
Surface chemistry	Biomolecule attachment
Fabrication route	Reproducibility and scalability

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Types of Metal Composites



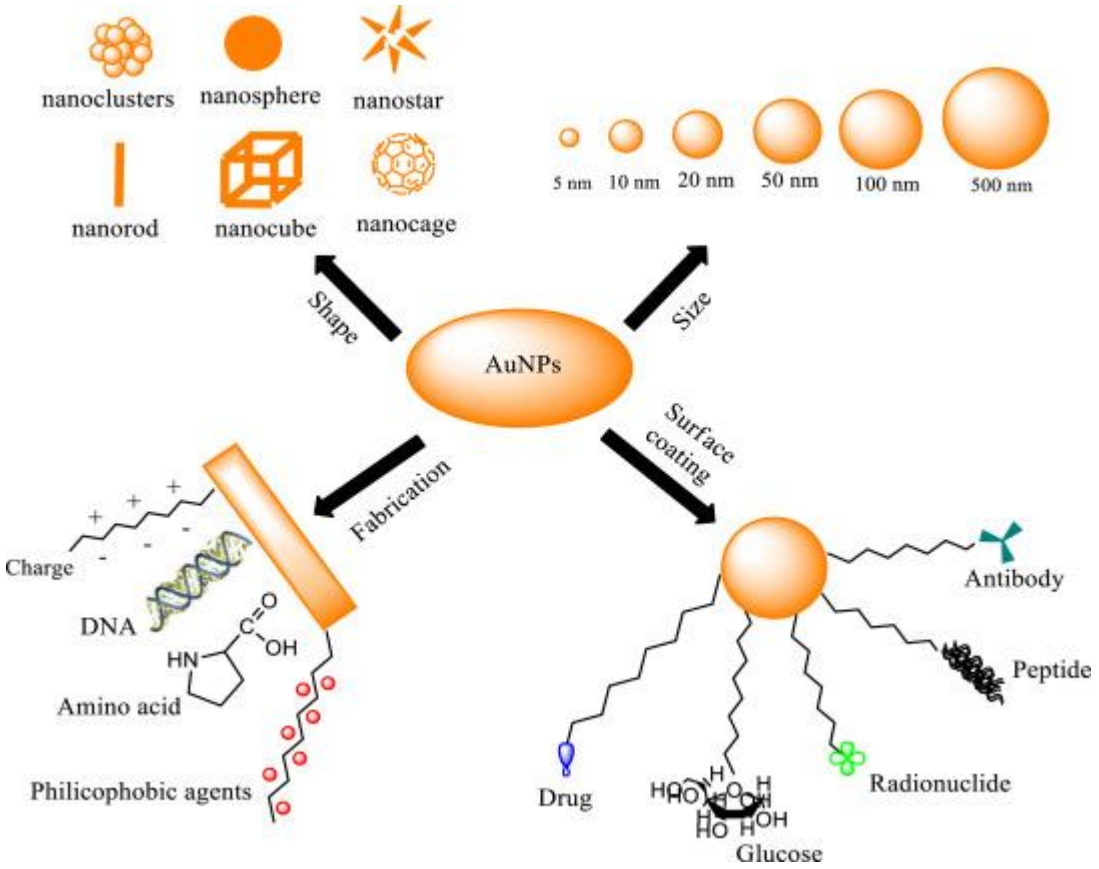


Same Composition + Different Architecture → Different Properties

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Why Au?

- ✓ Excellent chemical stability and corrosion resistance
- ✓ Strong localized surface plasmon resonance in the visible–NIR region
- ✓ Easy surface functionalisation through Au–thiol chemistry
- ✓ High electrical conductivity
- ✓ Excellent biocompatibility for biomedical devices
- ✓ Expensive



Measurement: Sensors 2021, 16, 100050

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Why Ag?

- Among the noble metals that exhibit plasmon resonance, silver (Ag) displays the highest efficiency of localized surface plasmon resonances (LSPR) excitation with a wider wavelength range. The LSPR band varies according to size, shape, composition and dielectric constant variations.
- The electrical, optical, catalytic and colour variations are tuned by inducing a change in morphology, size and aggregation behaviour.
- Ag NPs are more reactive and can easily undergo electrochemical oxidation due to their high extinction coefficients, larger scattering-to-extinction ratio and clear sharp bands.
- Ag possesses antimicrobial activity (bactericide properties, which can cause undesired effects such as DNA denaturation), vulnerable stability and toxicity.

Advantages

- Strongest plasmonic enhancement
- High scattering efficiency
- Excellent SERS performance
- Lower cost than Au

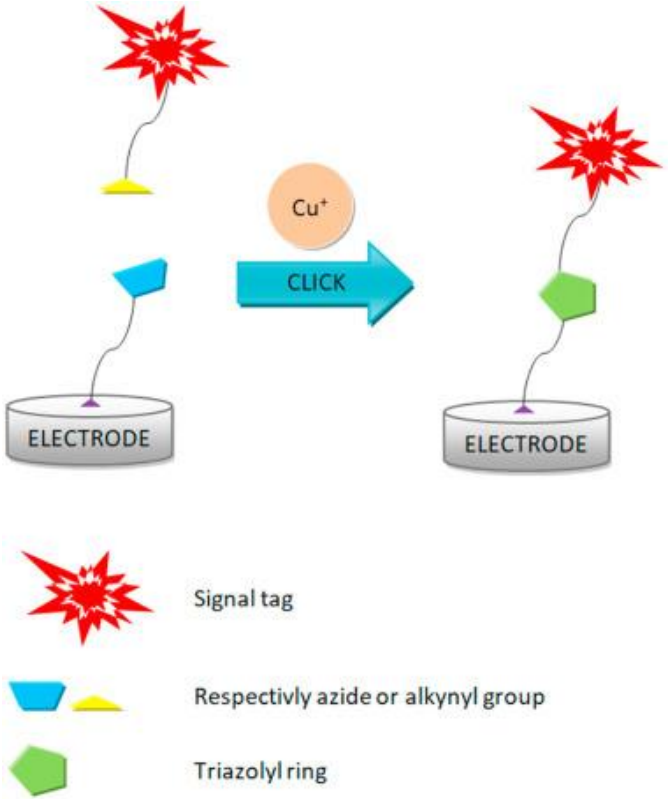
Limitations

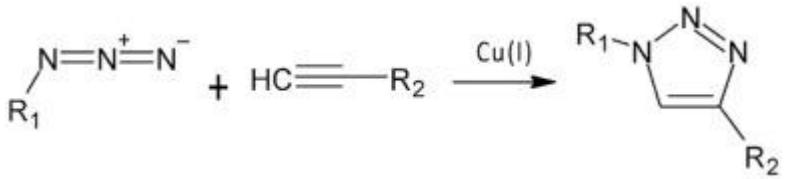
- Oxidation in ambient conditions
- Lower chemical stability
- Potential cytotoxicity
- Requires protective coating in biomedical devices

Sensors 2021, 21(15), 5114

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Why Cu?





General scheme for Cu^{I}-catalyzed azide/alkyne cycloaddition.

- ✓ **Copper is one of the most essential trace elements in organisms as it is involved in a wide range of metabolic processes.**
- ✓ **Non-precious metal Cu is earth-abundant, significantly cheaper and widely used in industries**
- ✓ **Cu possess unique catalytic and photoluminescence (PL) properties.**
- ✓ **Cu is popular for click Chemistry.**
- ✓ **Inexpensive**

Biosensors and Bioelectronics, 169, 2020, 112614

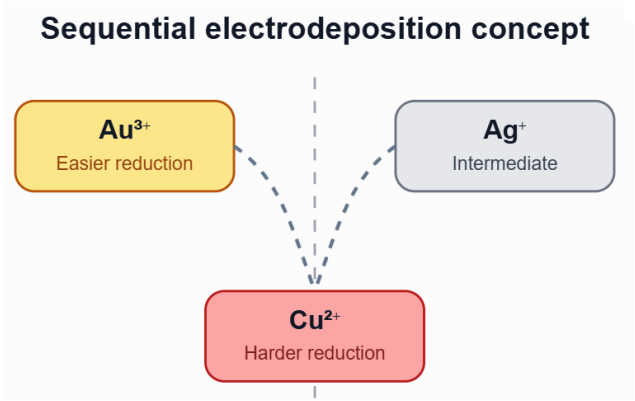
Why bimetallic or trimetallic over monometallic?

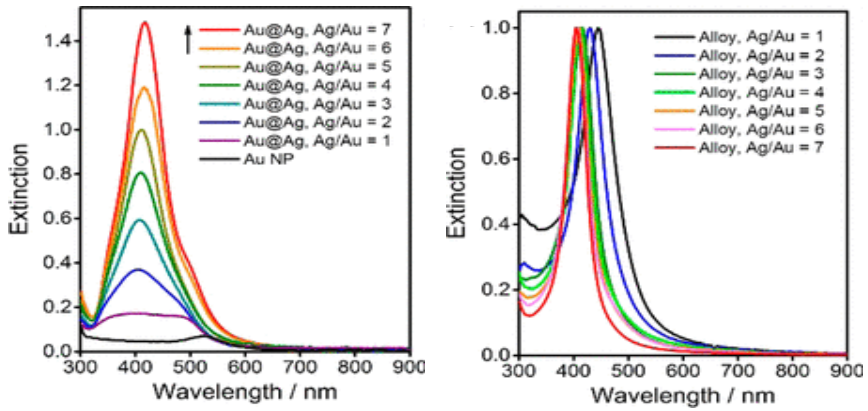
$Au^0 = +1.50$ eV

$Ag^0 = +0.80$ eV

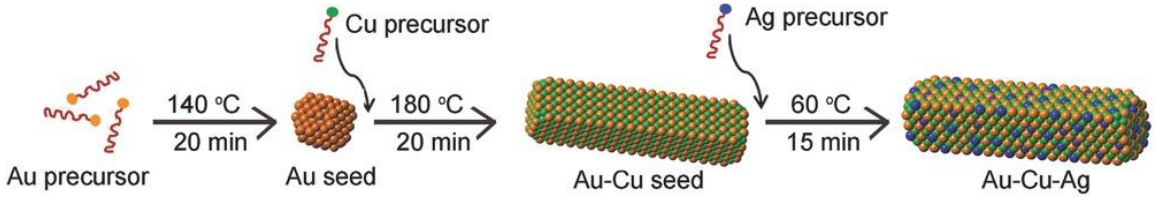
$Cu^0 = +0.34$ eV

Reduction potential of Au, Ag and Cu





Responses of Au/Ag system with their different ration



Schematic illustration of the synthetic strategy of Au–Cu–Ag trimetallic nanorods.

- Ag, Au, and Cu in a single unit to integrate the individual merits of each metal, the higher SERS activity of Ag, better surface stability and biocompatibility of Au, and structural stability of Cu.

- The reduction of Cu will be slower than Ag. Au will be more reductive than Ag. Therefore, the synthesis depends upon the sequence of metal, time gap, electrode potential and galvanic replacement of metal ion.

J. Am. Chem. Soc. 2014, 136, 20, 7474–7479
Plasmonic Particles, 34, 2017, 1600384

Exercise!

Q1. Q1. A bimetallic SERS substrate shows lower sensitivity than expected. Suggest three possible reasons related to its composition, structure or fabrication.

Q2. Suggest an Au–Ag–Cu tri-metallic nanocomposite design for biomedical applications (e.g., imaging, sensing, or therapy). Justify your choice based on the unique roles of each metal.

Q3. How could Material Informatics machine learning and AI contribute to optimizing the composition, synthesis, and performance prediction of multi-metallic nanocomposites?

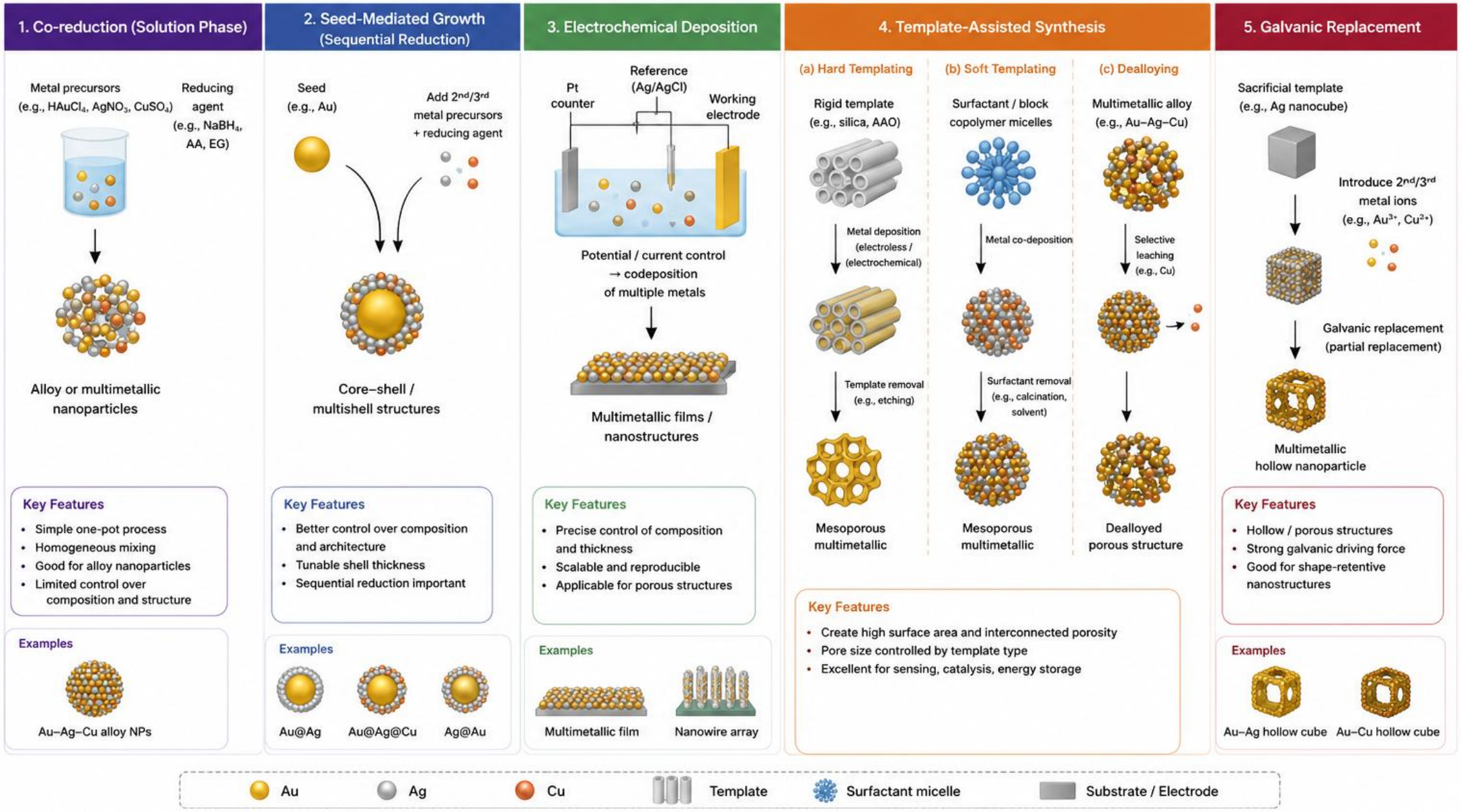
CRICOS code 00025B

Synthesis of bi and tr-metallic nanocomposite (mesoporous)

- **Hard-templating approach** (*i.e.*, nanocasting)
  - Hard templates made of inorganic materials such as mesoporous silica are commonly used to guide and control the building of mesoporous metals.
  - The generated replicas reverse the template's morphology; however, the dissolution process is typically slow and requires repeated exposure to dangerous chemicals.
  - Although hard templates provide a consistent and intuitive method for producing mesoporous metals, the complexity of the technique, lack of morphology control, and use of harsh chemicals have confined their use in wider applications
- **Soft-templating approach** (*e.g.*, lyotropic liquid crystalline (LLC) and di-block polymer-based templating)
  - Soft-templating methods are promising alternatives that use organic surfactants (*e.g.*, block polymers, lyotropic liquid crystals, *etc.*) as molecular templates to direct the formation of mesopores in metallic structures.
- **Template-free approach** (*e.g.*, post-synthetic dealloying)
  - The chemical treatment preferentially dissolves a non-targeted element, leaving a porous metal or alloy surface behind.
  - This technique is commonly seen in technologically significant alloy systems, such as stainless steel, brass, and different alloys especially Cu–Al alloys.
  - The mechanical and physical properties of a porous overlayer differ greatly from those of the bulk alloy on which it sits, resulting in stress corrosion cracking, brittle crack propagation as well as other unwanted material failure.

CRICOS code 00025B

Synthesis of bi and tr-metallic nanocomposite (mesoporous)



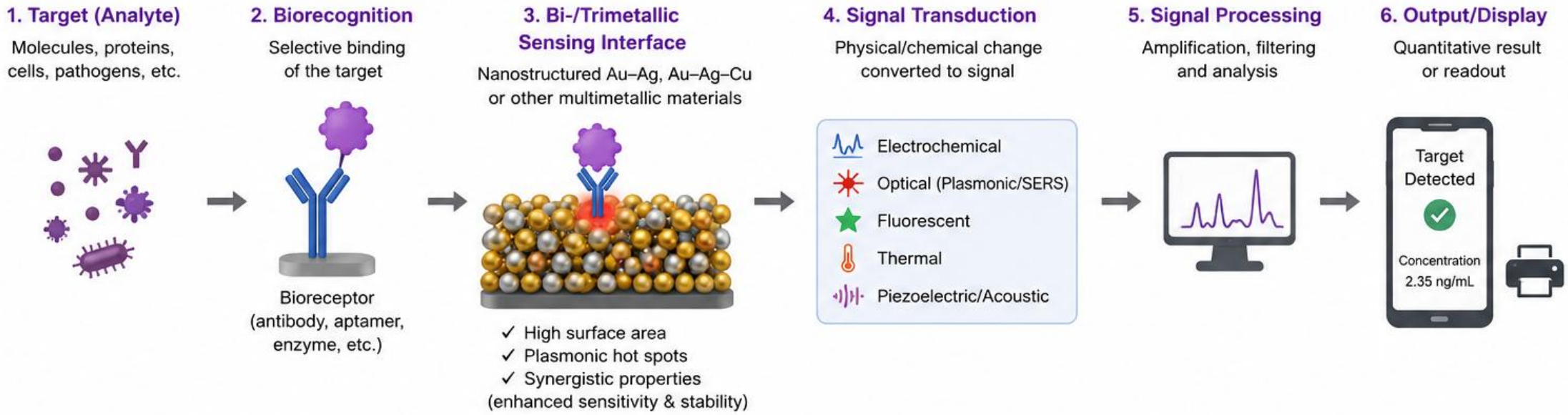
CRICOS code 00025B

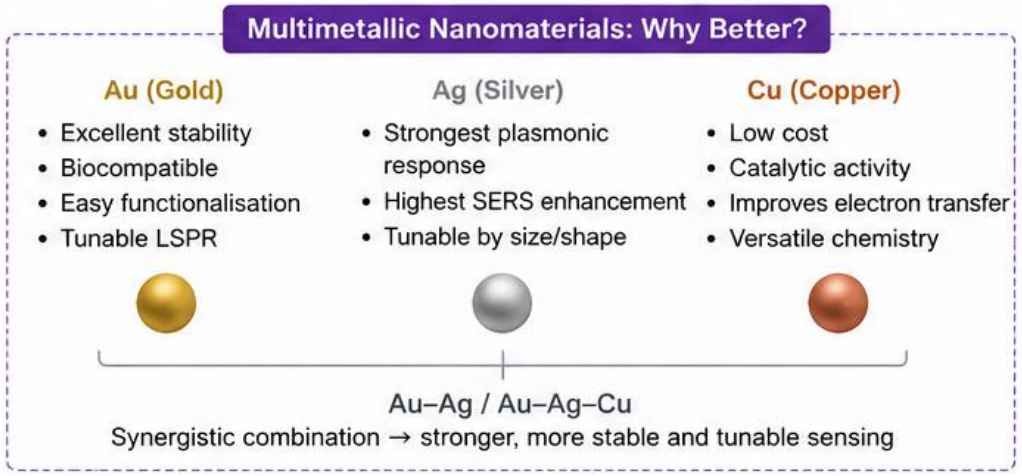
Synthesis of bi and tr-metallic nanocomposite (mesoporous)

Method	Advantage	Limitation	Engineering use
Hard templating	Excellent structural fidelity	Slow, harsh chemistry	Ordered porous metals
Soft templating	Tunable pores	Sensitive to processing	Mesoporous electrodes
Dealloying	Simple porous formation	Limited composition control	Porous alloys
Electrodeposition	Scalable and controllable	Requires conductive substrate	Sensor fabrication

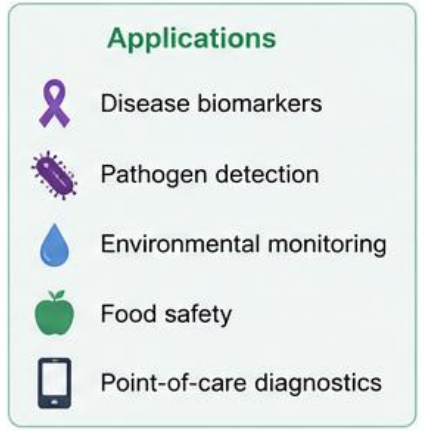
CRICOS code 00025B

Biosensor









Biosensors **2018**, *8*(4), 104

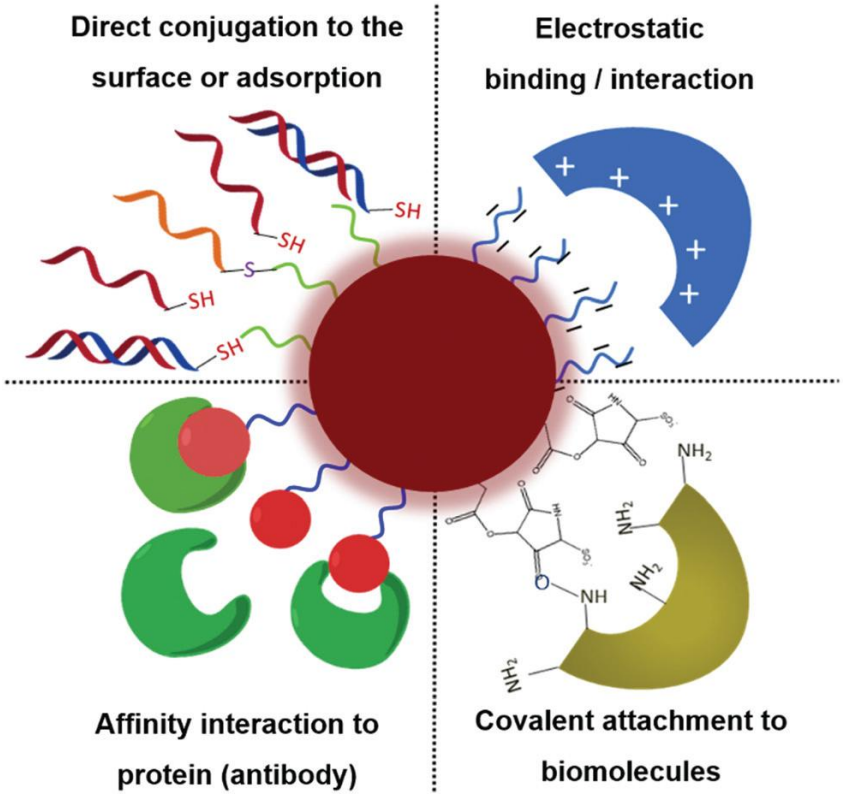
CRICOS code 00025B

Au, Ag and Cu in Biosensing : Surface Functionalization

- Their use in biosensing is inevitably about their high SA conferred greater functionalization abilities, making it possible to track any stimulus or material response in biological and non-biological configurations.

- The surface functionalization or covering of the surface with biocompatible compounds is crucial to (i) prevent degradation in an aqueous, acidic, or aggressive (blood) environment; (ii) suppress the magnetic interactions to avoid aggregation; and (iii) provide a functional group to bind or attach a wide range of biomolecules (e.g., proteins and nucleic acids).

- Notable functionalization mechanisms for tagging biological moieties to nanomaterial surfaces include;
  (i) thiol-NM surface interaction
  (ii) streptavidin-biotin interaction
  (iii) π-π stacking interactions, and
  (iv) NHS-EDC carbodiimide reaction.



Chem. Soc. Rev., 2019, 48, 5717

Surface Functionalization

Thiol–Au

Strong anchoring
SAM formation

EDC–NHS

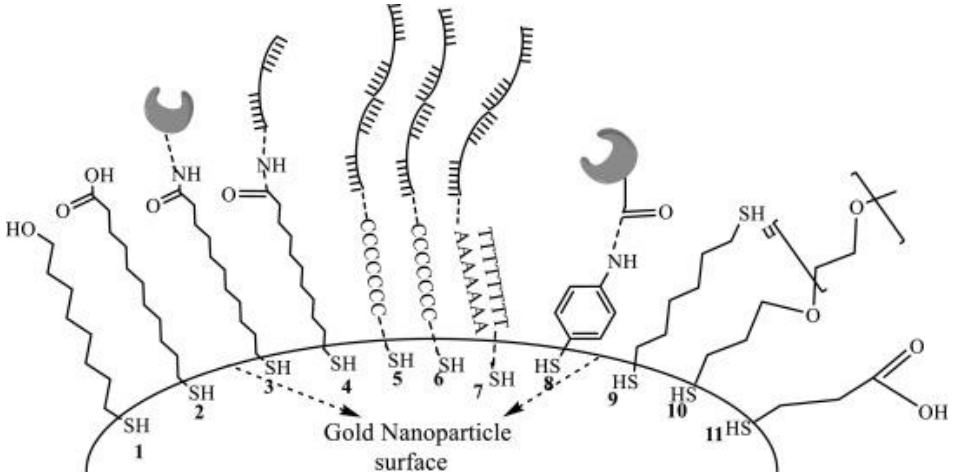
COOH + NH_2
Covalent amide bond

Streptavidin

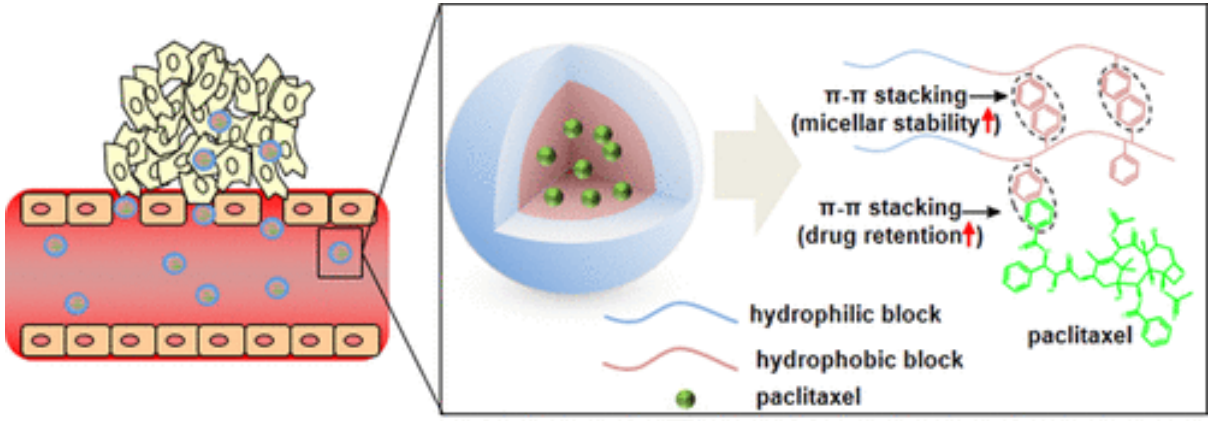
Biotin recognition
High affinity

π–π stacking

Aromatic surfaces
Non-covalent assembly



Modification with thiol compounds



Modification through π-π stacking interactions

Chem. Soc. Rev., 2019, 48, 5717

Exercise!

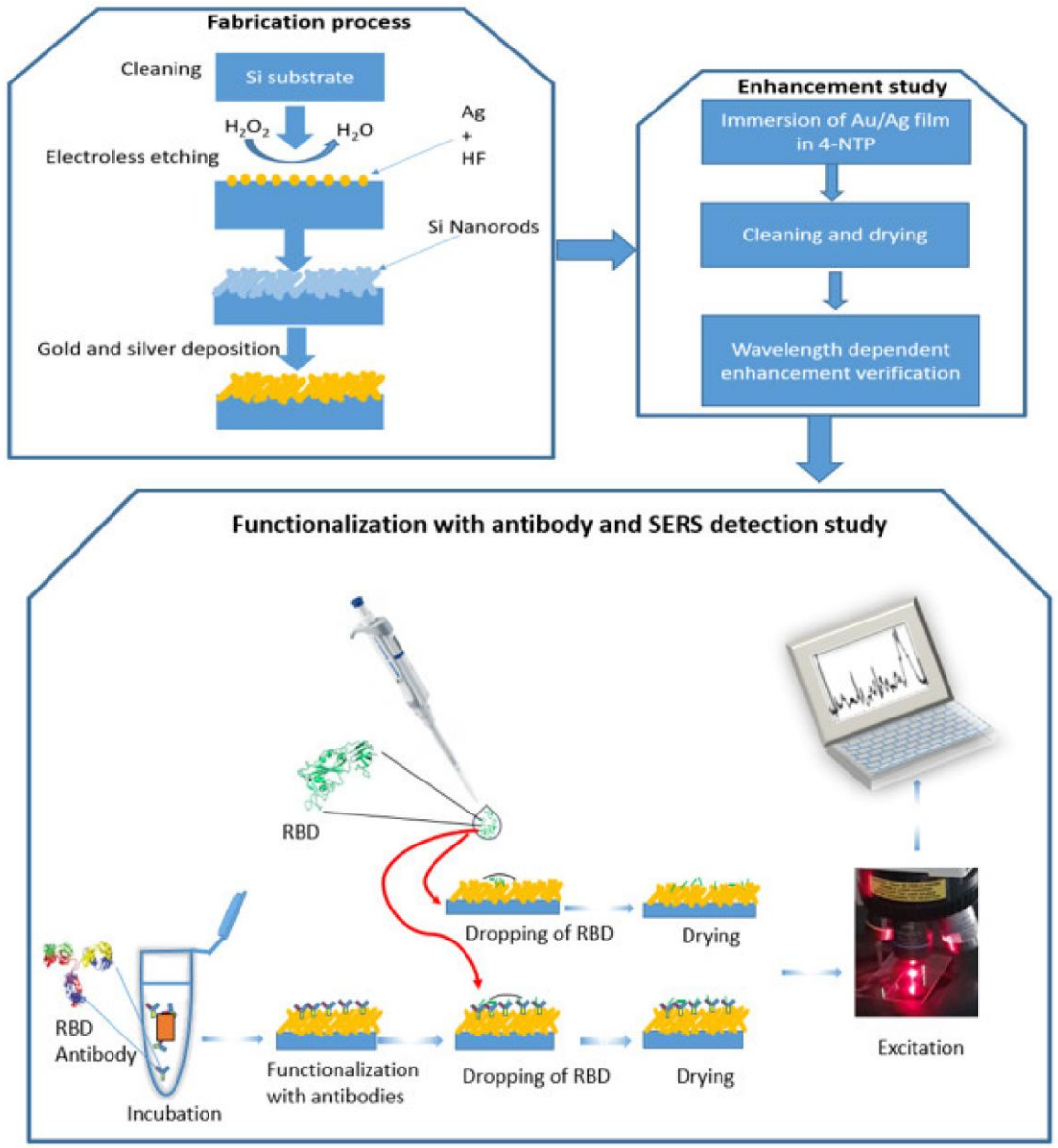
Q1. Functionalization improves stability, but may affect biocompatibility. How would you balance both for biomedical applications?

Q2. How can surface functionalization improve specificity of Au-Ag-Cu nanocomposites for detecting biomolecules?

Q3. A SERS sensor has excellent sensitivity but poor stability and reproducibility. How could surface functionalisation improve its performance?

CRICOS code 00025B

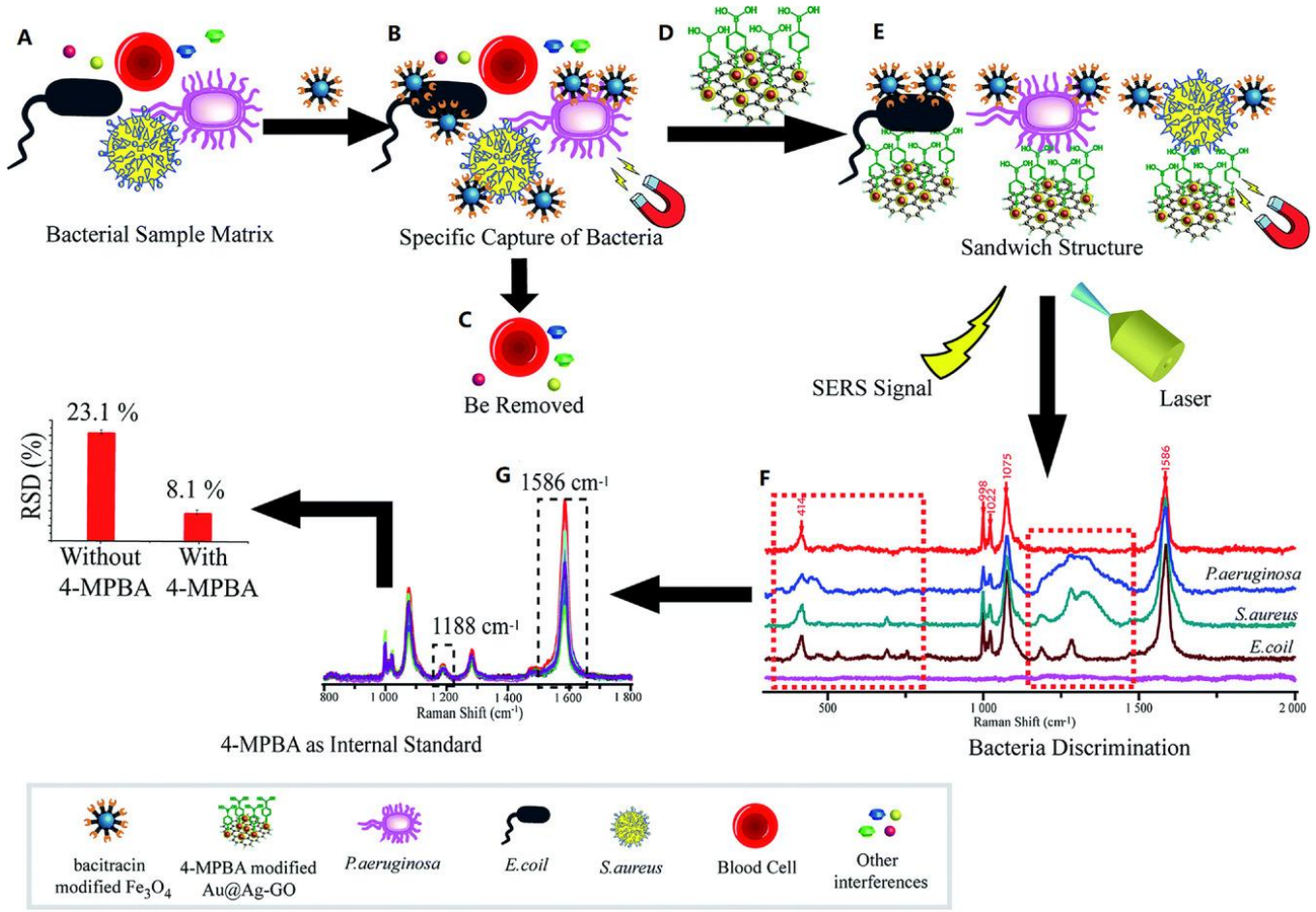
Au/Ag based Biosensor Platform: Viral detection



- ❑ What material was engineered?
- ❑ Why was Au/Ag chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

Sensors **2021**, *21*(13), 4617

Au/Ag based Biosensor Platform: Bacterial detection

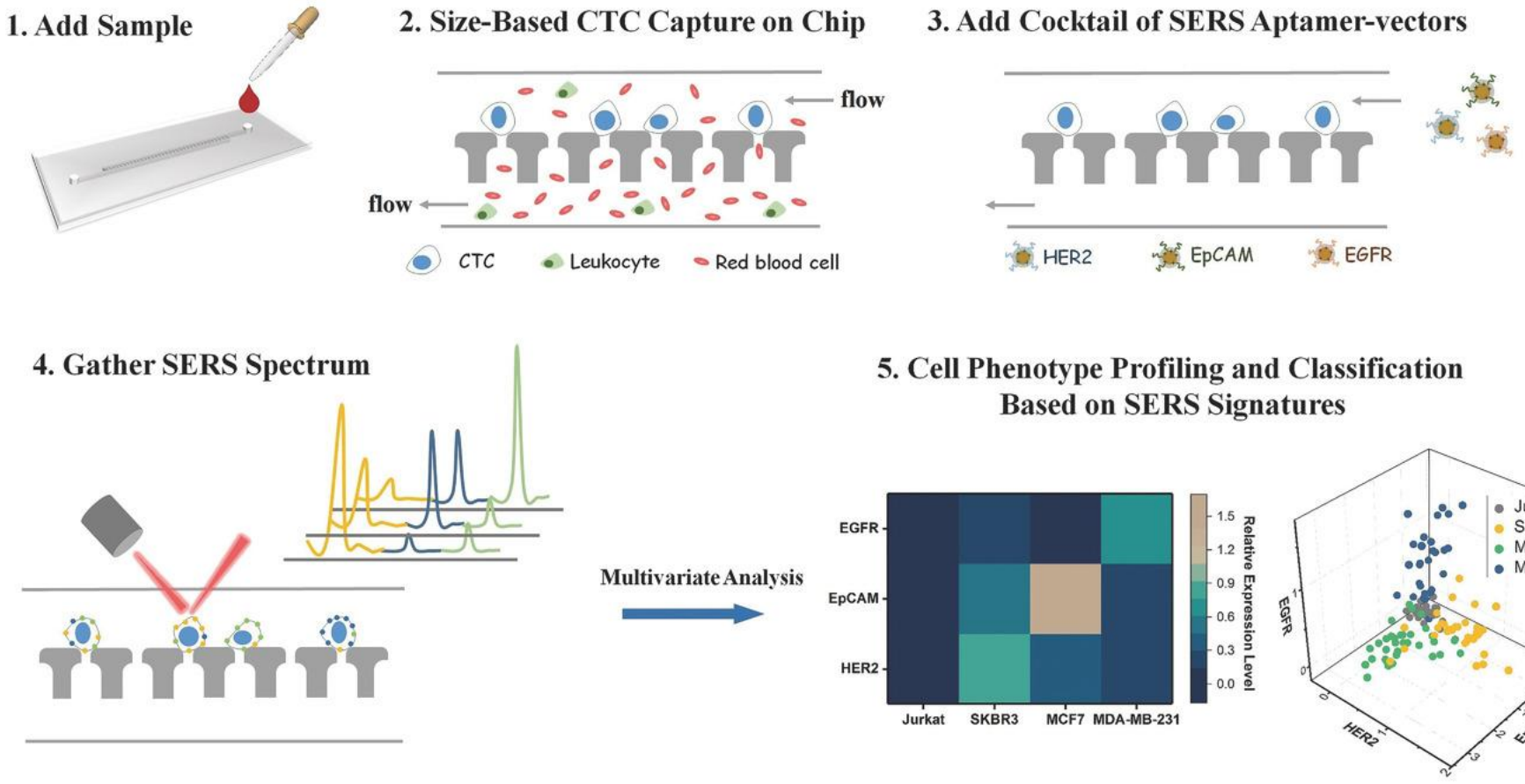


- ❑ What material was engineered?
- ❑ Why was Fe_2O_3 chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

Chem. Sci., 2018,**9**, 8781-8795

CRICOS code 00025B

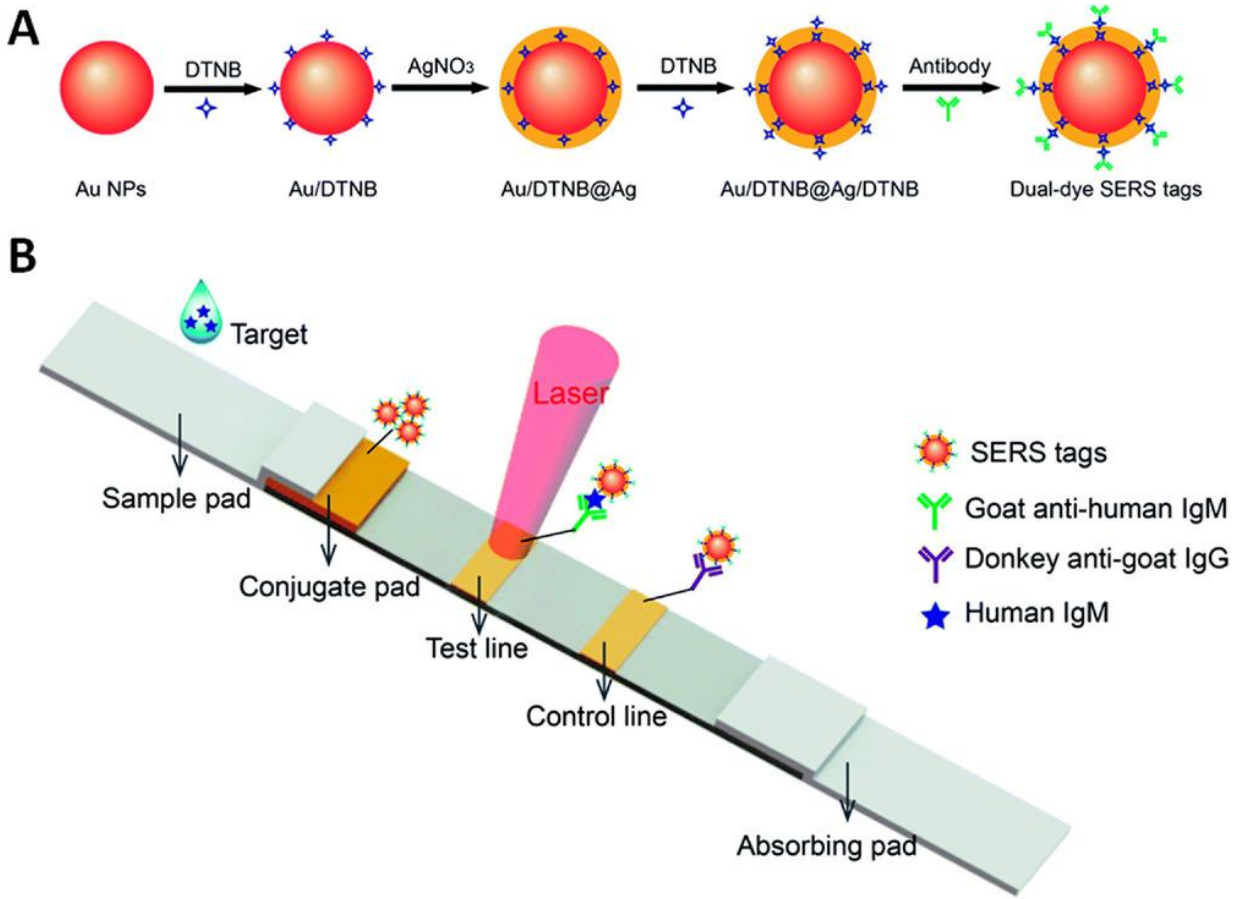
Au/Ag based Biosensor Platform: cancer cells detection



- ❑ Why use Au@Ag rather than Au alone?
- ❑ What role does the microfluidic structure play?
- ❑ How does multiplex SERS distinguish different biomarkers?

Small 2018, *17*, *1704433*

Au/Ag based Biosensor Platform: human IgM detection



- ❑ What material was engineered?
- ❑ Why was Au/Ag chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

RSC Adv. 2018, **8**, 21243.

Take-Home Messages

- ✓ **Composition matters: Combining metals can generate synergistic optical, catalytic and electronic properties.**
- ✓ **Architecture matters: Alloy, core–shell and porous structures can behave differently even with similar compositions.**
- ✓ **Processing matters: Synthesis controls composition, morphology, porosity and ultimately sensing performance.**
- ✓ **Engineering matters: The best sensor balances sensitivity, stability, reproducibility and manufacturability.**

Composition → Structure → Properties → Sensing Performance